
ProtSent: Protein Sentence Transformers

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Abstract

Protein language models (pLMs) produce per-residue representations that capture evolutionary and structural information, yet their mean-pooled sequence embeddings are not explicitly trained to reflect functional, evolutionary or structural similarity between proteins. We present Protein Sentence Transformers (ProtSent), a contrastive fine-tuning framework for adapting PLMs into general-purpose embedding models. ProtSent trains with MultipleNegativesRankingLoss across five protein-pair datasets: Pfam families, structurally derived hard negatives, AlphaFold DB structural pairs, and StringDB protein-protein interactions, and Deep Mutational Scanning data. We evaluate on 23 downstream tasks using frozen embeddings with a k -nearest-neighbor probe to measure embedding neighborhood quality. On ESM-2 150M, ProtSent improves 15 of 23 tasks, with gains of +105% on remote homology detection, +17% on variant effect prediction, and +19.9% Recall@1 on SCOPe-40 structural retrieval. The 35M variant improves 16 of 23 tasks with +40.5% on remote homology and +15.5% Recall@1 on SCOPe-40. Contrastive fine-tuning restructures the embedding space to better capture protein function and structure, without any task-specific supervision. We release the models, public data, and training recipe and code.

1 Introduction

In natural language processing, this analogous limitation of BERT-style models was addressed by Sentence-BERT [Reimers and Gurevych, 2019], which applies contrastive learning to restructure the embedding space so that semantically similar sentences become neighbors. Contrastive objectives optimize the metric that downstream tasks rely on: embedding proximity.

We apply this principle to protein language models. Our framework, Protein SentenceBERT (ProtSent), fine-tunes PLM backbones end-to-end using MultipleNegativesRankingLoss (MNRL) [Henderson et al., 2017] across multiple datasets, each capturing a different axis of biological relatedness. We combine five data sources: (i) Pfam family membership and (ii) structurally derived hard negatives, (iii) AlphaFold DB structural pairs, (iv) StringDB interaction pairs and (v) deep mutational scanning (DMS) data, using a CoSENT loss [Su, 2022] to capture continuous fitness landscapes.

We evaluate ProtSent on a suite of 23 downstream tasks using a deliberately simple protocol: embeddings are frozen and evaluated with a k -nearest-neighbor (KNN) probe. This evaluation strategy measures the quality of the embedding space geometry, instead of the capacity of a learned

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classifier on top of it. The rationale is that if contrastive training successfully restructures the neighborhood structure, a probe that relies entirely on neighbor identity should be the most sensitive detector of improvement.

Our contributions are as follows:

- We introduce ProtSent, a contrastive fine-tuning framework for protein language models that combines five protein-pair datasets with round-robin sampling.
- We demonstrate that contrastive fine-tuning produces substantial gains on tasks that depend on embedding neighborhood quality, including +105% on remote homology detection and +19.9% Recall@1 on SCOPe-40 structural retrieval (ESM-2 150M), with similar gains at the 35M scale (+40.5% and +15.5% respectively).
- We show that these improvements are consistent across two model scales (35M and 150M parameters) and multiple biological tasks spanning function, structure, engineering and mutation.
- We provide ablation studies isolating the contribution of each training data source and the sampling strategy to the final performance.
- We release the models², code³, public data, and training recipe.

2 Related work

Protein language models. Self-supervised protein language models learn residue-level representations from large sequence databases. ESM-1b and ESM-2 [Rives et al., 2021, Lin et al., 2023] train Transformer encoders with masked language modeling on millions of UniRef sequences, producing embeddings that encode evolutionary conservation, secondary structure, and contact information. ESMFold [Lin et al., 2023] extended this approach to atomic-resolution structure prediction. ProtTrans [Elnaggar et al., 2022] explored several architectures (BERT, Albert, T5) at billion-parameter scale and showed that per-residue representations transfer well to secondary-structure and localization tasks. TAPE [Rao et al., 2019] introduced a standardized benchmark suite and demonstrated that pretrained representations improve over hand-crafted features across five protein tasks. A common theme is that downstream transfer is typically mediated by a learned probe, a linear layer or small MLP trained on task-specific labels. In contrast, ProtSent targets the sequence-level embedding space, asking whether contrastive fine-tuning can restructure it so that nearest-neighbor lookup alone is sufficient for multiple downstream tasks.

Contrastive fine-tuning of language models. In NLP, Sentence-BERT [Reimers and Gurevych, 2019] showed that fine-tuning BERT with a siamese objective on natural-language inference pairs produces sentence embeddings whose cosine similarity correlates with semantic similarity, reducing the computational cost of sentence comparison from quadratic cross-encoder inference to a single embedding lookup. Subsequent work extended this to multilingual and domain-specific settings [Reimers and Gurevych, 2020]. For proteins, Heinzinger et al. [2022] proposed ProtTucker, which fine-tunes ProtT5 with triplet loss on CATH superfamily labels (S30 subset, 3,186 training proteins) and demonstrated improved remote-homology detection. Independently, Redl et al. [2023] adapted the SentenceTransformers framework to ProtBERT and showed that contrastive training on disorder and stability annotations improves performance on those specific tasks. Our work differs from both in scale and breadth: ProtSent trains on over 70 million protein pairs drawn from five heterogeneous data sources (Pfam families, structural pairs, interaction networks, hard negatives, and DMS), and evaluates on 23 tasks spanning classification, regression, and retrieval, rather than a single axis of similarity. We aim to present a foundation model for universal use.

Embedding-based protein search and retrieval. Traditional sequence search tools such as BLAST and HMMer rely on local alignment heuristics and profile Hidden Markov Models [Söding, 2005, Steinegger and Söding, 2017]. Recent works have explored learned embeddings as an alternative. PLMSearch [Liu et al., 2024] combines ESM-1b embeddings with a lightweight cross-attention module to re-rank homology search results, achieving higher sensitivity than HMMer on remote

²<https://huggingface.co/collections/oriel9p/protsent>

³<https://github.com/oriel9p/ProtSent>

homologs. Hong et al. [2024] introduced a dense homolog retrieval (DHR) system that trains a dual-encoder with contrastive learning on SCOPe domains and reports 93% sensitivity at 1% false-positive rate. Foldseek [van Kempen et al., 2024] takes a different approach, encoding 3D backbone geometry as a structural alphabet and performing fast structural search without full alignment. These systems are optimized end-to-end for retrieval; ProtSent instead targets general-purpose embeddings that transfer across diverse tasks, with retrieval (SCOPe-40) serving as one evaluation axis among many.

Protein function prediction. Predicting enzyme class, fitness effects, or other functional properties from sequence remains a core challenge. GOBeacon [Lin et al., 2025] recently proposed an ensemble framework that applies contrastive regularization to multi-label GO classifiers, reporting improvements on CAFA-style benchmarks [Jiang et al., 2016]. Unsupervised and anomaly-based approaches do not necessarily beat supervised methods [Michael-Pitschaze et al., 2024, Ofer and Linial, 2025]. A related subproblem is variant-effect prediction, where supervised approaches train on deep mutational scanning (DMS) assays and zero-shot methods leverage pLM log-likelihoods [Rives et al., 2021]. ProtSent takes a middle path: it uses DMS data during training as an auxiliary regression signal (CoSENT loss) but does not train a task-specific predictor, instead relying on the restructured embedding space to capture continuous fitness landscapes.

3 Method

3.1 Problem formulation

Let $f_\theta : \mathcal{S} \rightarrow \mathbb{R}^d$ denote a protein language model parameterized by θ that maps an amino acid sequence $s \in \mathcal{S}$ to a d -dimensional embedding. Given a set of protein pairs (s_i, s_j) drawn from biological relationships (shared family, structural similarity, physical interaction), our goal is to fine-tune θ so that the cosine similarity $\text{sim}(f_\theta(s_i), f_\theta(s_j))$ is high for biologically related pairs and low for unrelated pairs. After fine-tuning, the embedding space should support downstream tasks through simple nearest-neighbor lookup, without task-specific supervision.

3.2 Backbone architecture

We use ESM-2 [Lin et al., 2023] as the backbone encoder, wrapped in the SentenceTransformers framework [Reimers and Gurevych, 2019]. ESM-2 is a protein language model pretrained with masked language modeling on UniRef50 sequences [Suzek et al., 2015]. We experiment with two model scales: ESM-2 35M (12 layers, 480-dimensional embeddings) and ESM-2 150M (30 layers, 640-dimensional embeddings). During contrastive fine-tuning, all Transformer layers are fine-tuned. Sequence-level embeddings are obtained by mean-pooling over non-padding residue tokens, producing a single vector $\mathbf{h} \in \mathbb{R}^d$ per protein. Input sequences are truncated to first 512 residues; we do not apply center cropping.

3.3 Training data

ProtSent trains on five protein-pair datasets summarized in table Table 1. Further details and filtering parameters are in Appendix 12:

Table 1: Training datasets after all filtering, clustering, and decontamination (raw upstream sizes are larger; see Appendix 12). “Group label” is the positive-pair criterion for MNRL sources; STRING is pair-native and DMS uses continuous CoSENT scores.

Dataset (post-filter)	Rows / pairs	Group label	Loss
Pfam (linclust@70%)	32.9M domains	Pfam family ID	MNRL
Pfam Hard Negatives	1.8M anchors (1.2M with HN)	Pfam family ID + explicit neg.	MNRL
AFDB50, Foldseek-grouped	133.9M sequences	Foldseek structural cluster	MNRL
STRING-DB v12 PPI	36.5M pairs	Pair-native	MNRL
ProteinGym DMS / clinical	2.2M pairs	Continuous score	CoSENT

Pfam family pairs. Pfam-A full-alignment domains. Positives are pairs belonging to the same family [Mistry et al., 2021]. We deduplicate with MMseqs2 easy-linclust at 70% identity, leaving

32.9M domains in 26,796 families and 15,284 clans. In-batch negatives are other families within each batch. Singletons families are dropped. This dataset encodes evolutionary and functional homology at the domain level.

Pfam hard negatives. We generate novel hard evolutionary negatives. For each anchor per family, using the Pfam-A Hidden Markov Model match-state emissions (PSSM), we sample mutants with ≥ 3 point substitutions at distinct positions spaced at least $\max(6, L/8)$ apart, drawn from positions with per-residue $\Delta S < -1.0$, until the total log-odds drop satisfies $\sum_i \Delta S_i \leq -16.0$. This is an arbitrary cutoff for being likely to "break the conserved profile" and be deleterious, while retaining $>98\%$ similarity. These hard negatives force the model to learn fine-grained discriminative features beyond simple sequence identity.

AlphaFold DB structural pairs. AFDB50 sequences (pLDDT > 70 , non-fragment), from AlphaFold DB [Varadi et al., 2022, Barrio-Hernandez et al., 2023] predicted structures are joined on their AFDB50 representative with the Steinegger-Lab AFDB Foldseek [van Kempen et al., 2024] structural clusters (cluster flags $\{1, 2\}$). The positive label is the Foldseek cluster. This provides large scale structural supervision, forcing the embedding space to capture three-dimensional structural relationships. These are unsupervised, predicted clusters (unlike curated scop families).

StringDB interaction pairs. Protein-protein interaction pairs from the STRING database [Szklarczyk et al., 2023], filtered to `combined_score` ≥ 400 (medium confidence). These capture functional association, co-occurrence in pathways, complexes or co-expression, providing a distinct biological relationship axis from homology or structural similarity. Sequences with over 50% sequence identity, by MMseqs2 linclust, to any protein in the Bernett PPI benchmark test set were removed from the StringDB data to prevent data leakage. Remaining sequences were globally deduplicated to 50% identity, and filtered for lengths $[10, 1024]$.

DMS fitness data. ProteinGym DMS and clinical substitutions and indels scores are per-assay z-scored, clipped to $[-3, 3]$ and rescaled to $[0, 1]$; clinical labels mapped Pathogenic $\rightarrow 0$, Benign $\rightarrow 1$. This auxiliary loss operates on single proteins rather than pairs. Assay families that overlap downstream evaluation (GB1 and GFP variants) are dropped, and the supervised-benchmark test fold is removed using its explicit split metadata where present, or the same deterministic per-group 80/20 split otherwise, so no DMS row used in training is later evaluated as supervised regression.

Leakage controls. The primary risk of train-test overlap arises from the Pfam and AlphaFold DB datasets. For Pfam, training pairs are sampled from family membership labels, which do not overlap with the held-out evaluation splits used for individual downstream tasks (e.g., the remote homology fold-level evaluation uses a disjoint fold partition, not family-level labels). For AlphaFold, training uses Foldseek cluster co-membership rather than SCOPe labels; we do not filter AFDB sequences against SCOPe test domains, so partial sequence overlap is possible. We note this as a limitation: while the training labels (Foldseek clusters) and evaluation labels (SCOPe superfamilies) are drawn from different classification systems, the underlying sequences may overlap. For DMS data we exclude ProteinGym from our evaluation due to overlap. For the remaining evaluation tasks (solubility, signal peptide, etc.), the training data provides only pairwise relationship labels, not task-specific annotations.

3.4 Training objective

The primary contrastive objective is MultipleNegativesRankingLoss (MNRL) [Henderson et al., 2017], defined over mini-batches of positive pairs. For a batch of N anchor-positive pairs $\{(s_i, s_j)\}_{i=1}^N$, the loss treats all other positives in the batch as negatives:

$$\mathcal{L}_{\text{MNRL}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{sim}(f_{\theta}(s_i), f_{\theta}(s_j))/\tau)}{\sum_{j=1}^N \exp(\text{sim}(f_{\theta}(s_i), f_{\theta}(s_j^+))/\tau)} \quad (1)$$

where τ is a temperature parameter (we use $\tau = 0.05$, equivalently a scale factor of 20) and $\text{sim}(\cdot, \cdot)$ denotes cosine similarity.

For the Pfam hard negatives dataset, explicit negative columns are appended to the batch, providing additional challenging negatives beyond the in-batch negatives.

The auxiliary DMS loss uses CoSENTLoss [Su, 2022], a ranking-based objective that preserves the ordering of continuous fitness scores among protein variants. The DMS dataset is included as an additional entry in the round-robin sampler (Section 3.5), receiving equal per-step weight with the four MNRL datasets. At each training step, the sampler draws a batch from exactly one dataset; thus the DMS loss contributes to approximately one-fifth of all gradient updates rather than being weighted by an explicit coefficient λ .

3.5 Multi-dataset sampling

The datasets are combined using round-robin sampling: at each training step, the dataloader draws a batch from the next dataset in a cyclic order. This ensures equal representation of each biological signal source regardless of dataset size, preventing the largest dataset from dominating the gradient updates.

3.6 Training configuration

Models were trained on a single NVIDIA RTX 6000 Ada GPU (48 GB). We use AdamW with a cosine learning rate schedule, 0.1 dropout and an effective batch size of 1024. Up to 70M training pairs are generated from the source datasets. Full hyperparameters are listed in Table 6 (Appendix 8).

4 Experiments

4.1 Evaluation protocol

We evaluate all models using a frozen embedding protocol: the fine-tuned (or baseline) model encodes each protein sequence into a fixed-length vector, and a k -nearest-neighbor (KNN) probe with $k=3$ and Euclidean distance is trained on the resulting embeddings. Although training optimizes cosine similarity, for approximately L2-normalized embeddings (as produced by contrastive training), Euclidean distance and cosine distance induce the same neighbor ranking: $\|a-b\|^2 = 2(1-\cos(a,b))$ when $\|a\| = \|b\| = 1$. We use the scikit-learn default (Euclidean) for compatibility; the SCOPe-40 retrieval evaluation (Section 5.2) uses cosine distance and shows similar results.

For binary and multiclass classification tasks, we report Area Under the ROC Curve (AUC). For multiclass tasks, we report macro-averaged F1 score. For regression tasks, we report Spearman rank correlation. Evaluation uses a held-out test split where available, with 4-fold cross-validation on the training set as a fallback.

4.2 Benchmark tasks

We evaluate on 23 tasks in three types:

Binary classification (8 tasks). Protein-protein interaction prediction (Bernett Gold Standard PPI dataset), solubility prediction (DeepSol [Khurana et al., 2018]), peptide-HLA binding, metal ion binding, signal peptide prediction (SignalP [Teufel et al., 2022]), neuropeptide precursor prediction (NeuroPID [Ofer and Linial, 2015]), binary subcellular localization, and material production classification.

Multiclass classification (5 tasks). Remote homology detection at the fold level, enzyme commission (EC) number classification, subcellular localization (10-class), antibiotic resistance mechanism classification, and temperature stability classification (thermophile/mesophile/psychrophile).

Regression (10 tasks). Variant effect prediction (GB1 fitness landscape), fluorescence prediction (TAPE benchmark [Rao et al., 2019]), protein stability (Biomap), thermostability (FLIP benchmark [Dallago et al., 2021]), optimal pH prediction, enzyme catalytic efficiency (kcat), cloning classification, beta-lactamase fitness (PEER benchmark), AAV capsid fitness (FLIP), and RhIA enzyme mutation effects.

4.3 SCOPe-40 structural retrieval

In addition to the probe-based evaluation, we evaluate embedding quality through a retrieval task on the SCOPe-40 structural classification database. For each query protein, we retrieve the nearest neighbors by cosine similarity in embedding space and measure Recall@ K for $K \in \{1, 10, 30\}$, where a retrieval is considered correct if the retrieved protein belongs to the same structural superfamily as the query.

5 Results

5.1 KNN probe evaluation

Table 2 presents the KNN probe evaluation across all 23 tasks for both the 35M and 150M backbones. ProtSent improves 16 of 23 tasks at the 35M scale and 15 of 23 at 150M, with the largest gains on tasks that depend on structural or functional grouping in embedding space.

At the 35M scale, the strongest improvements appear on remote homology detection (+40.5%), RhlA enzyme mutations (+77.2%), beta-lactamase fitness (+18.5%), and fluorescence (+15.6%). These tasks require the model to group proteins by structural fold or mutational phenotype, contrastive training on Pfam families and AlphaFold structural pairs is designed to capture this relationship. Binary classification tasks improve broadly: PPI (+5.3%), peptide-HLA binding (+3.6%), and solubility (+2.7%).

The 150M backbone exhibits the same pattern with several notable differences. Remote homology detection reaches +105%, and EC classification improves by +15.9% (vs. +0.3% at 35M), showing the larger backbone provides enough capacity for contrastive training to separate enzyme classes. GB1 variant effect prediction improves by +17.3% whereas the 35M model showed a slight decrease (−0.8%), indicating that capturing mutational fitness relationships benefits from additional model capacity. Conversely, RhlA degrades at 150M (−27.1%) despite its large gain at 35M, and beta-lactamase shows only +0.1% at 150M vs. +18.5% at 35M.

Stability and thermostability regression degrade modestly at both scales, possibly because the sequence level representation does not specifically preserve ordinal relationships within a protein’s mutational neighborhood in the case of point mutations.

5.2 SCOPe-40 structural retrieval

Table 3 reports retrieval performance on the SCOPe-40 structural classification benchmark for both model scales. ProtSent achieves consistent double-digit improvements across all recall thresholds, demonstrating that contrastive fine-tuning meaningfully reorganizes the embedding space to place structurally similar proteins closer together.

The improvements are similar across both model scales and all recall thresholds. The 150M model shows even larger gains (+19.9% at Recall@1) than the 35M (+15.5%), indicating the larger backbone provides more capacity for contrastive fine-tuning to reorganize the embedding space. The single nearest neighbor in the ProtSent embedding space is substantially more likely to share the query protein’s structural fold compared to the baseline ESM-2 representation.

5.3 Ablation studies

We conduct six ablation experiments on the ESM-2 35M backbone. Each ablation removes one data source, we also replace round-robin with proportional sampling, retraining from scratch with otherwise identical hyperparameters.

Removing Pfam family pairs causes the largest degradation, reducing the number of improved tasks from 16 to 15 and the mean improvement from +6.7% to +4.6% (Table 4). Individual tasks that depend on evolutionary homology are most affected: beta-lactamase fitness drops −8.8% and RhlA enzyme mutations drops −31.5% relative to the full model. This confirms that Pfam provides the dominant contrastive signal.

Training without the hard negatives dataset yields a surprising result: 20 of 23 tasks improve over the baseline, compared to 16 for the full model, with a higher mean delta (+7.9% vs. +6.7%)

Table 2: KNN probe evaluation on 23 downstream tasks: baseline ESM-2 vs. ProtSent at two model scales (35M and 150M parameters). $\Delta\% = 100 \times (\text{Trained} - \text{Baseline})/\text{Baseline}$. Positive $\Delta\%$ values are **bolded**. Tasks are grouped by type.

Task	Metric	ESM-2 35M			ESM-2 150M		
		Base	ProtSent	$\Delta\%$	Base	ProtSent	$\Delta\%$
<i>Multiclass classification</i>							
Remote Homology (Fold)	F1 _M	.223	.313	+40.5	.190	.390	+105.0
EC Classification	F1 _M	.450	.452	+0.3	.408	.473	+15.9
Subcellular Localisation	AUC	.784	.794	+1.3	.797	.813	+1.9
Antibiotic Resistance	AUC	.785	.786	+0.1	.786	.790	+0.6
Temperature Stability	F1 _M	.896	.873	−2.6	.906	.881	−2.8
<i>Binary classification</i>							
PPI (Bernett)	AUC	.560	.589	+5.3	.556	.592	+6.4
Peptide-HLA Binding	AUC	.748	.775	+3.6	.749	.772	+3.1
Solubility (DeepSol)	AUC	.692	.711	+2.7	.712	.719	+1.1
Neuropeptide (NeuroPID)	AUC	.948	.956	+0.9	.963	.945	−1.8
Binary Subcel. Loc.	AUC	.925	.932	+0.8	.936	.922	−1.5
Signal Peptide	AUC	.965	.971	+0.7	.971	.972	+0.1
Metal Ion Binding	AUC	.825	.830	+0.5	.808	.843	+4.3
Material Production	AUC	.755	.758	+0.4	.754	.759	+0.6
<i>Regression</i>							
RhIA Enzyme Mutations	Spr.	.236	.418	+77.2	.345	.252	−27.1
Beta-lactamase (PEER)	Spr.	.670	.793	+18.5	.732	.733	+0.1
Fluorescence (TAPE)	Spr.	.490	.567	+15.6	.504	.569	+12.7
Variant Effect (GB1)	Spr.	.656	.651	−0.8	.670	.785	+17.3
Optimal pH	Spr.	.498	.514	+3.1	.509	.512	+0.6
AAV Fitness (FLIP)	Spr.	.742	.729	−1.7	.706	.725	+2.6
Cloning Classification	Spr.	.398	.394	−1.0	.386	.389	+0.8
Enzyme Cat. Efficiency	Spr.	.705	.687	−2.6	.705	.699	−0.9
Stability (Biomap)	Spr.	.568	.547	−3.7	.588	.569	−3.3
Thermostability (FLIP)	Spr.	.475	.449	−5.3	.467	.460	−1.4
Tasks improved		16/23			15/23		

Table 3: SCOPe-40 structural retrieval results. Recall@ K measures the fraction of queries for which a protein from the same structural superfamily appears among the top- K nearest neighbors.

Model	Recall@1	Recall@10	Recall@30
ESM-2 35M (baseline)	0.385	0.588	0.641
ProtSent 35M	0.445	0.651	0.710
$\Delta\%$	+15.5	+10.7	+10.9
ESM-2 150M (baseline)	0.423	0.589	0.644
ProtSent 150M	0.507	0.685	0.724
$\Delta\%$	+19.9	+16.4	+12.3

(Table 7). This suggests that explicit hard negative mining, while potentially useful for fine-grained discrimination, may introduce overly aggressive contrastive gradients that perturb the embedding space on some tasks. EC classification improves by +7.4% relative to the full model when hard negatives are removed. We hypothesize that the in-batch negatives alone provide sufficient contrastive signal for most downstream tasks, and that hard negative selection warrants further investigation.

Removing AlphaFold DB structural pairs causes the second-largest degradation after Pfam: the number of improved tasks drops from 16 to 13 and the mean delta falls from +6.7% to +3.2%. EC classification is hit hardest (−11.0% vs. baseline), and remote homology drops to +15.3% (from +40.5% with the full model), confirming that structural supervision from Foldseek clusters provides a complementary signal that Pfam family labels alone do not fully capture. Removing StringDB has a milder effect: 17 tasks still improve and the mean delta matches the full model (+5.9%). PPI prediction drops to −0.5% (from +5.3%), confirming that StringDB interaction pairs drive the PPI improvement. However, the overall embedding quality remains largely intact, suggesting the other three data sources provide sufficient contrastive signal for most tasks.

Replacing round-robin with proportional sampling yields 16 improved tasks with a mean delta of +7.0%, comparable to the full model. The modest difference indicates that the training is relatively

Table 4: Ablation study results (ESM-2 35M, KNN probe). Each row shows the number of tasks improved (out of 23) and the mean relative change (%) across all tasks, both measured against the stock ESM-2 baseline. The full model row corresponds to the default ProtSent configuration.

Configuration	Tasks improved (/23)	Mean $\Delta\%$
Full model (ProtSent)	16	+6.7
w/o Pfam families	15	+4.6
w/o hard negatives	20	+7.9
w/o AlphaFold DB	13	+3.2
w/o StringDB	17	+5.9
w/o DMS (CoSENT)	15	+5.8
Proportional sampling	16	+7.0

Table 5: Few-shot evaluation (ESM-2 35M, KNN probe). Relative improvement ($\Delta\%$) of ProtSent over baseline at each sample size. Bold indicates improvement.

Task	Type	$N=50$	$N=100$	$N=500$	$N=1000$
Remote Homology	Multi.	N/A	+244.5	+97.8	+90.2
Subcellular Loc.	Multi.	+29.5	+19.7	+20.9	+5.6
EC Classification	Multi.	-61.5	+64.9	+39.1	+14.1
Fluorescence	Reg.	+7.0	+7.8	+8.7	+27.8
Peptide-HLA	Bin.	+2.0	+3.3	+8.2	+1.1
Metal Ion Binding	Bin.	-0.2	+17.1	+0.9	-1.0
Solubility	Bin.	+7.5	+14.3	-5.2	-0.4
Neuropeptide	Bin.	-0.9	+1.9	+1.9	+2.2
Signal Peptide	Bin.	-3.6	+1.8	+2.3	+1.8
PPI (Bernett)	Bin.	-24.0	-5.1	+3.7	+1.9
Enzyme Cat. Eff.	Reg.	-126.9	-22.1	+4.0	+18.0
Beta-lactamase	Reg.	+30.2	-62.7	-8.8	+2.4
Variant Effect (GB1)	Reg.	-67.3	-40.2	-0.5	-2.3
Stability (Biomap)	Reg.	-53.2	-49.9	-7.4	-3.4
Tasks improved		6/14	10/14	10/14	10/14

robust to the sampling strategy, though individual task variation exists (e.g., optimal pH drops -9.4% vs. the full model under proportional sampling).

Removing the DMS CoSENT loss modestly reduces performance (15/23 tasks, +5.8% vs. 16/23, +6.7%). Fitness-regression tasks are most affected: fluorescence drops from +15.6% to +10.4% and thermostability degrades further. The auxiliary DMS signal thus provides a small but steady benefit to fitness-landscape tasks without harming broader embedding quality. Detailed per task changes per ablation are provided in the appendix. These are single-factor ablations; interactions between components remain unexplored and could yield additional insights.

5.4 Few-shot evaluation

A key motivation better embedding neighborhoods is improved performance under label scarcity. We evaluate ProtSent in a few-shot setting by subsampling the training set to $N \in \{50, 100, 500, 1000\}$ labeled examples per task and evaluating with the KNN probe ($k=3$). Table 5 reports the relative improvement ($\Delta\%$) of ProtSent over the baseline ESM-2 35M at each sample budget.

At $N \geq 100$, ProtSent improves 10 of 14 evaluable tasks. The largest gains appear on tasks that require grouping by structural or functional similarity: remote homology detection (+244.5% at $N=100$, +90.2% at $N=1000$), subcellular localisation (+5.6-29.5%), and EC classification (+14.1-64.9%). These tasks benefit most from improved neighborhood structure because the KNN probe relies on finding correctly labeled neighbors in a small training set. At $N=50$, results are noisy: with $k=3$ and only 50 training points, the probe is highly sensitive to stochastic neighbor selection, and several regression tasks show large negative deltas that stabilize at higher budgets. Notably, tasks where ProtSent degrades in the few-shot regime (Variant Effect, Stability) are the same tasks that show

modest degradation with the full dataset, possibly representing a trade-off of the contrastive objective rather than a few-shot-specific failure mode.

6 Discussion

Neighborhood structure as the primary improvement axis. The consistent pattern across both model scales is that tasks requiring accurate embedding neighborhoods, remote homology, structural retrieval, fluorescence prediction, show the largest improvements. Contrastive training optimizes angular relationships between embeddings, restructuring neighborhoods to reflect the biological similarities encoded in the training data. Tasks where the baseline already performed well (e.g., signal peptide, temperature stability) show smaller or negative changes, hinting the global reorganization of the space can disrupt incidental structure that the pretrained model had learned.

Task-specific analysis. The +105% improvement on remote homology detection (150M, KNN) and +40.5% (35M, KNN) is the most dramatic result. Fold-level homology requires recognizing structural similarity between proteins with minimal sequence identity, precisely the capability that contrastive training on Pfam families and AlphaFold structural pairs should confer. The improvement on PPI prediction (+5–6% across scales) confirms that interaction partners are brought closer in embedding space by training on StringDB pairs.

Regression tasks. Fluorescence and beta-lactamase fitness improve at both scales, indicating that contrastive training preserves local manifold structure relevant to mutational effects. GB1 variant effect prediction presents a scale-dependent result: -0.8% at 35M but $+17.3\%$ at 150M, indicating the larger backbone allows the contrastive objective to capture fitness landscape relationships without disrupting other properties. Some regression tasks (thermostability, stability) degrade at both scales, possibly because the contrastive objective does not specifically preserve ordinal relationships within a single protein’s mutational neighborhood. ProtSent is not designed to compete with specialized variant-effect prediction methods that use supervised heads or retrieval-augmented inference; instead, the VEP results characterize a side effect of general-purpose embedding improvement on a task that was not directly optimized.

Limitations. The training data also limited to the relationship types captured by the five source datasets; protein relationships not represented (e.g., enzymatic mechanism similarity, expression patterns) are not specifically optimized. We do not compare to specialized retrieval systems (ProtTucker, PLMSearch, DHR) on matched benchmarks; our goal is general-purpose embeddings rather than retrieval-optimized models, but such comparisons would help quantify the generality–accuracy trade-off. Finally, all results are from single training runs without multi-seed uncertainty estimates; the few-shot results in particular show high variance at small N , and confidence intervals would strengthen the robustness claims.

7 Conclusion

We presented ProtSent, a contrastive fine-tuning framework that transforms protein language model representations into function-aware embeddings where proximity reflects biological relatedness. By training with multiple data sources and round-robin sampling, ProtSent achieves substantial improvements on tasks that depend on embedding neighborhood quality. The uniform improvements across two model scales suggest that contrastive fine-tuning is a broadly applicable strategy for adapting pretrained protein language models to produce embeddings suitable for retrieval, clustering, and few-shot transfer.

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8 Additional training details

Training was conducted on NVIDIA RTX 6000 Ada GPUs (48 GB VRAM) on an HPC cluster. The 35M model trains in approximately 3–4 hours; the 150M model trains in approximately 1.3 days. We use the SentenceTransformers library [Reimers and Gurevych, 2019] built on top of HuggingFace Transformers [Wolf et al., 2020]. All models are trained for a single epoch over 70M generated pairs. Full hyperparameters are listed in Table 6.

Table 6: Training hyperparameters for both model scales.

Hyperparameter	ESM-2 35M	ESM-2 150M
Per-device batch size	64	16
Gradient accumulation	16	64
Effective batch size	1024	1024
Learning rate	3×10^{-4}	2×10^{-4}
Warmup steps	500	1000
LR scheduler	Cosine with min LR	
Max sequence length	512	
Optimizer	AdamW (fused)	
Dropout	0.1	
Max training pairs	70M	
Epochs	1	
Multi-dataset sampler	Round-robin	

9 Full benchmark task descriptions

Below we provide additional details for selected evaluation tasks. Metrics and probe types for all 23 tasks are specified in Section 4.1 and the results tables.

Remote homology detection. Fold-level classification from the SCOPe database. Training and test sets are split by superfamily so that no superfamily appears in both; the model must recognize structural similarity across evolutionary distant sequences. We report macro-averaged F1 across fold classes.

SCOPe-40 structural retrieval. A retrieval task over SCOPe domains filtered at 40% sequence identity. For each query, we retrieve nearest neighbors by cosine similarity and evaluate Recall@ K for $K \in \{1, 10, 30\}$, where a hit is correct if it shares the query’s structural superfamily. This task uses the full validation set (100,000 proteins) with no subsampling.

Variant effect prediction (GB1). Regression on the GB1 protein fitness landscape, where each variant is scored by its experimentally measured binding affinity. We report Spearman rank correlation between predicted and true fitness values. The GB1 landscape contains $\sim 150,000$ single and multi-site variants.

Other tasks. The remaining tasks follow standard formulations from the TAPE [Rao et al., 2019], FLIP [Dallago et al., 2021], and PEER benchmark suites. Binary classification tasks (PPI, solubility, signal peptide, metal ion binding, peptide-HLA, neuropeptide, binary subcellular localization, material production) are evaluated by AUC. Multiclass tasks (EC number, subcellular localization, antibiotic resistance, temperature stability) use AUC or macro F1. Regression tasks (fluorescence, stability, thermostability, optimal pH, enzyme catalytic efficiency, cloning, beta-lactamase, AAV, RhIA) use Spearman correlation. All tasks use held-out validation splits where available, with 4-fold cross-validation as a fallback.

10 UMAP Visualization of Embedding Space

Figure 1 shows UMAP projections of protein embeddings from the baseline ESM-2 and ProtSent models, computed on held-out SCOPe-40 domains colored by fold-level and superfamily-level

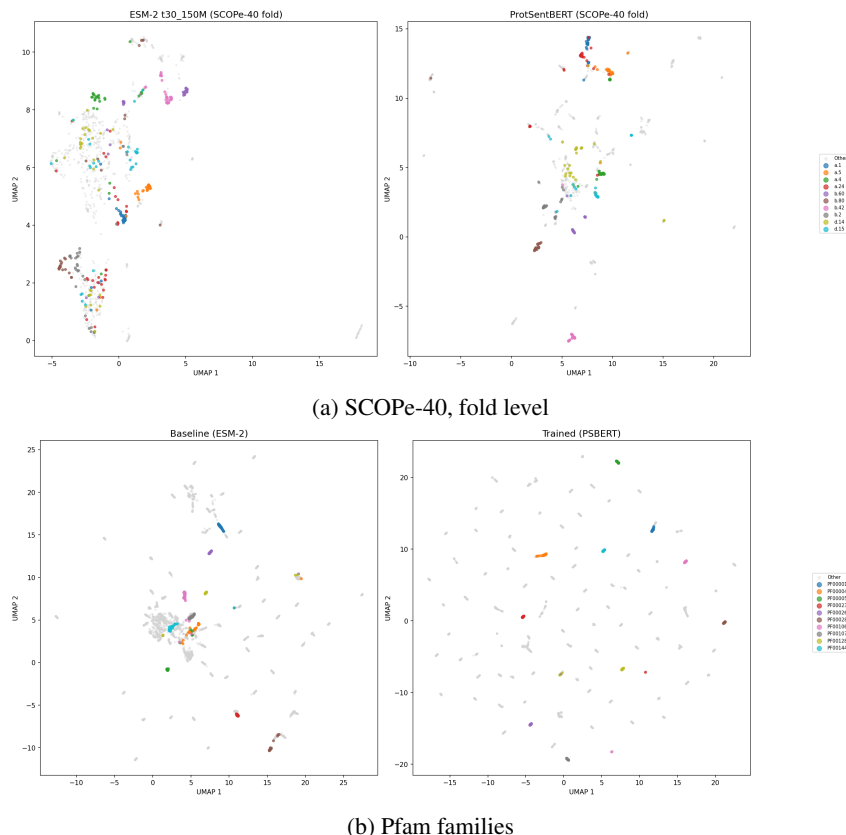


Figure 1: UMAP projections of baseline ESM-2 150M (left in each panel) vs. ProtSent 150M (right). Points are colored by the 10 most frequent groups; remaining groups in gray. Scopr-40 at the superfamily level, and Pfam families.

labels, as well as on Pfam sequences colored by family. Across all three groupings and both model scales, the ProtSent embeddings exhibit visually tighter and more separated clusters compared to the baseline. The effect is most pronounced at the fold level, where baseline embeddings show substantial overlap between structural classes that is reduced after contrastive fine-tuning. We note that UMAP projections are sensitive to hyperparameters and do not constitute a quantitative evaluation; these visualizations are intended as a qualitative complement to the retrieval and probe results.

11 Full Per-Task Ablation Results

Table 7 provides the complete per-task breakdown of all ablation experiments. Each cell reports the relative change (%) of the ablated model compared to the stock ESM-2 35M baseline, using a KNN probe (except multilabel tasks which require a linear probe). The summary row counts tasks improved and reports the mean relative delta across the 23 KNN-evaluated tasks.

12 Dataset construction details

Each training parquet is reproduced by `python data_prep.py -dataset X` at default flags (with one exception, noted below for AFDB).

12.1 Pfam (pfam_sorted.parquet)

Source. Pfam-A.fasta.gz and Pfam-A.clans.tsv.gz from ftp.ebi.ac.uk/pub/databases/Pfam/current_release. Headers parse to (domain_id, family_id, sequence) with the PFxxxxx version stripped.

Table 7: Per-task ablation results (ESM-2 35M, KNN probe except multilabel tasks which use linear probe). Each cell shows the relative change (%) vs. the stock ESM-2 baseline. Bold indicates best configuration per task.

Task	Metric	Full	−Pfam	−HardNeg	−AFDB	−StringDB	Prop.	−DMS
<i>Binary</i>								
PPI	AUC	+5.3	+5.9	+4.3	+5.5	−0.5	+6.6	+5.9
Peptide-HLA	AUC	+3.6	+4.0	+7.0	+4.0	+4.9	+6.2	+5.0
Solubility	AUC	+2.7	+2.5	+2.6	+2.8	+1.3	+2.8	+2.4
Neuropeptide	AUC	+0.9	+0.1	+1.1	+0.1	+0.1	+0.8	+0.2
Binary SubCell	AUC	+0.8	+1.0	+0.9	+0.7	+0.8	+1.4	+0.8
Signal Peptide	AUC	+0.7	+1.0	+1.2	−0.3	+0.1	+1.4	−0.1
Metal Ion	AUC	+0.5	+1.6	+1.7	−1.8	+0.7	+0.9	+0.7
Material Prod.	AUC	+0.4	+0.8	+0.6	−0.3	+0.7	+1.1	+1.2
<i>Multiclass</i>								
Remote Homology	F1	+40.5	+42.5	+49.5	+15.3	+43.3	+54.7	+40.6
SubCell Loc.	AUC	+1.3	+1.8	+2.2	+1.5	−3.1	+1.6	+2.0
EC Number	F1	+0.3	−1.3	+7.7	−11.1	+3.7	+2.5	−2.1
Antibiotic Res.	AUC	+0.1	−0.2	+0.0	+0.2	+0.7	+0.6	−0.3
Temp. Stability	F1	−2.6	−1.1	−1.3	−1.2	−9.3	−1.2	−1.9
<i>Multilabel (linear probe)</i>								
CAFA5	F1	−2.1	−1.7	−1.1	−6.7	−5.6	+0.6	−1.2
Mol. Function	F1	−4.6	−4.1	−2.4	−7.2	−5.0	+1.0	−3.3
<i>Regression</i>								
RhIA	Sp.	+77.2	+21.3	+48.6	+38.0	+36.2	+59.3	+59.3
Beta-lactamase	Sp.	+18.5	+8.1	+17.8	+7.2	+19.5	+9.3	+8.9
Fluorescence	Sp.	+15.6	+20.6	+24.2	+22.0	+19.6	+18.4	+10.4
Optimal pH	Sp.	+3.1	−3.7	+5.9	−6.0	+4.7	−6.6	+2.2
Variant (GB1)	Sp.	−0.8	+9.0	+7.9	+11.8	+13.8	+8.4	+14.7
Cloning	Sp.	−1.0	−1.4	−1.2	−6.9	−0.6	−1.4	−0.9
AAV Fitness	Sp.	−1.7	+0.2	+3.6	+1.2	−1.5	−0.2	−1.1
Enzyme Eff.	Sp.	−2.6	−1.8	+0.1	−3.7	+1.3	−0.1	+0.0
Stability	Sp.	−3.7	−5.4	+1.0	−2.2	−5.8	−2.5	−3.2
Thermostability	Sp.	−5.3	−0.6	−4.4	−3.3	+5.2	−2.1	−11.8
Tasks improved		16/23	15/23	20/23	13/23	17/23	16/23	15/23
Mean $\Delta\%$		+6.7	+4.6	+7.9	+3.2	+5.9	+7.0	+5.8

Pipeline. (1) MMseqs2 easy-linclust -min-seq-id 0.7 -cov-mode 1 -c 0.8 on the raw Pfam-A FASTA (global asymmetric clustering; family members assigned to representatives from other families are dropped). (2) Left-join family → clan; orphan families inherit clan_id := family_id. (3) Drop singleton families. (4) Sort by (clan_id, family_id) so windowed slicing preserves clan-level diversity during training.

Final. 32,943,498 sequences, 26,796 families, 15,284 clans (input: ~62M raw Pfam-A FASTA records).

12.2 Pfam hard negatives (pfam_hard_negatives.parquet)

Source. pfam_sorted.parquet above plus Pfam-A.hmm.gz (parsed with pyhmmer).

PSSM. For each Pfam HMM, take match-state emissions $e_{i,a}$ and the pyhmmer amino-acid background b_a (both clipped at 10^{-9}), set $S_{i,a} = \log_2 e_{i,a} - \log_2 b_a$.

Eligibility. Drop families whose median sequence length differs from the HMM model length by more than 10%. Per row: keep $5 < L < 1024$; cap at 100 sequences/family.

Sampling. For an anchor of length L , use direct map $i \rightarrow i$ for $i < \min(L, \text{model_len})$; compute $\Delta S_{i,a} = S_{i,a} - S_{i,\text{wt}(i)}$ and zero-out self-substitutions and non-standard residues. The candidate pool is positions (i, a) with $\Delta S_{i,a} < -1.0$. Rejection-sample k -tuples (per-family seed $42 + i$, up to 2,048 proposals per k) with $k_{\min} = \max(6, \lceil -16.0 / \min_i \Delta S_i^{(\min)} \rceil)$ and $k_{\max} \leq 50$, accepting the first proposal that satisfies $\sum_i \Delta S_i \leq -16.0$ and pairwise position spacing $\geq \max(6, L/8)$.

Final. 1,821,068 anchors over 24,047 families (of 26,796 in pfam_sorted, after the $\pm 10\%$ length filter and the per-family cap of 100); 1,210,230 anchors (66%) carry a non-null hard_negative.

12.3 AFDB Foldseek structural pairs (afdb_sorted.parquet)

Sources. (i) willdaspt/afdb_clustered_seqs (HuggingFace) — AFDB sequences with their AFDB50 representative IDs, plus per-row plddt and fragment flags. (ii) 1-AFDBClusters-repId_entryId_cluFlag_taxId.tsv.gz from afdb-cluster.steineggerlab.workers.dev/v6/ — the upstream Foldseek S-cluster mapping. We retain cluFlag $\in \{1, 2\}$ (singleton and small-cluster flags are excluded).

Pipeline. (1) Lazy-scan HF parquets, filter plddt > 70 and fragment = 0. (2) Inner-join on HF.repId == Steinegger.entry_id; carry the Foldseek representative as group_id and the AFDB50 representative as afdb50_cluster_id. (3) Pre-shuffle (seed 42) then sort by afdb50_cluster_id. (4) Run uncapped: we use -limit_gb 0 (the CLI default of 25 GB caps at ~50M rows).

Final. 133,856,004 sequences in 815,712 Foldseek structural clusters spanning 1,815,626 AFDB50 representatives (input: ~214M AFDB sequences upstream; the pLDDT > 70 + non-fragment filter and the cluFlag $\in \{1, 2\}$ inner-join produce the drop).

Disambiguation. “AFDB”, “AFDB50”, and “AFDB Foldseek clusters” name three different objects: the full AlphaFold Database; AFDB sequences clustered at 50% identity (the AFDB50 release); and AFDB structural clusters from Foldseek over predicted structures (the v6 S-cluster file). The contrastive positive label is the third; the underlying sequence resource is the second.

12.4 STRING-DB PPI (stringdb_train.parquet)

Sources. STRING v12.0 protein.sequences.v12.0.fa.gz and protein.physical.links.full.v12.0.txt.gz from stringdb-downloads.org.

Pipeline. (1) Pre-filter the FASTA to proteins that appear in some link with combined_score ≥ 400 (STRING medium-confidence; reduces ~59M $\rightarrow$ ~25M proteins). (2) Bernett decontamination: load the Synthyra/bernett_gold_ppi test split, write each test sequence under a BERNETT_* ID, run a single MMseqs2 easy-linclust at 50% identity / 80% target coverage on the union, and drop any STRING protein whose linclust cluster contains any BERNETT_* member. (3) Two-stage clustering of survivors (all -cov-mode 1): easy-linclust at 65%/85% then cascaded cluster at 50%/75% on stage-1 reps, with mappings composed member $\rightarrow$ rep₆₅ $\rightarrow$ rep₅₀. (4) Pair construction: filter links by score, join both endpoints to rep-50 clusters, drop self-cluster edges, canonicalise unordered cluster pairs, sort by score descending, deduplicate keeping the highest-scoring edge per cluster pair. (5) Length filter [10, 1024] on both endpoints; final shuffle (seed 42); write only (seq1, seq2).

Final. 36,502,692 pairs over 6.73M unique proteins per endpoint (input: STRING v12.0 has ~59M proteins and >20B physical links; the score ≥ 400 filter, Bernett decontam, and cluster-pair deduplication account for the reduction).

12.5 ProteinGym DMS / clinical (dms_cosent.parquet)

Source. OATML-Markslab/ProteinGym_v1 on HuggingFace, splits DMS_substitutions, DMS_indels, clinical_substitutions, clinical_indels.

Pipeline. (1) Continuous DMS: per-assay z-score of DMS_score clipped to $[-3, 3]$ and rescaled to $[0, 1]$. (2) Clinical: Pathogenic $\rightarrow$ 0.0, Benign $\rightarrow$ 1.0. (3) Drop assays with DMS_id starting GB1_ or GFP_AEQVI_ (overlap with our protein-level benchmarks). (4) Test-fold drop: if a recognised split column (stage, split, set, fold) carries train/test labels, drop test rows; otherwise apply the supervised benchmark’s deterministic per-group 80/20 split (RandomState(42), groups of DMS_id for DMS rows or protein_id for clinical rows). Groups with fewer than 10 rows are not split (the supervised benchmark also skips these at evaluation, so this introduces no leak). Pairs that appear in any test row are also removed globally as a final pass. (5) Final shuffle (seed 42). Optional mutant-mutant intra-assay pairing is disabled.

Final. 2,175,734 pairs over 3,576 unique wild-type targets and 2.09M unique mutants (input: ~3M raw ProteinGym v1 rows across the four splits; the GB1/GFP assay drop and the supervised-test-fold drop account for the reduction). CoSENT preserves the score-induced ordering of pairs rather than training to absolute targets.

12.6 Compute and reproducibility

End-to-end on a single 48-thread workstation: STRING two-stage clustering under 6 h, AFDB join ~ 30 min on NVMe, Pfam hard-negative generation $\sim 2\text{k}$ anchors/s on 32 threads. All thresholds, identity cutoffs, and seeds are fixed in `data_prep.py`; re-running with default flags (plus `-limit_gb 0` for AFDB) reproduces every parquet bit-for-bit against a fixed snapshot of the upstream HuggingFace, EBI, and STRING-DB releases.